

MACHINE LEARNING BASED EARLY WARNING SYSTEM FOR RISK MANAGEMENT

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ABSTRACT

This report presents an early warning stock risk management system developed during a four-month internship in the domain of Data Science. The project addresses the problem of quantifying risk for a company by combining market-based signals with unstructured news and article sentiment. The objective is to generate operational risk alerts that are both numerically grounded and qualitatively informed, while avoiding common pitfalls such as look-ahead bias in financial time series modeling.

The proposed approach integrates two complementary data streams: (i) internal market data (OHLCV price history from yfinance library), from which risk factors such as volatility, Value at Risk (VaR) and Sharpe ratio are derived (ii) external data in the form of recent public news, which is analysed using Google Gemini to extract sentiment and a qualitative risk profile. A logistic regression model, trained with a strictly time-based train-test split and strong regularization, maps the market-derived features to discrete risk levels (low, medium, high).

Experimental results on Amazon (AMZN) data from 2010-2023 show that the model achieves an accuracy of 0.808, precision of 0.918, recall of 0.808 and F1-score of 0.835 on the testing data. The integration of Gemini-based news sentiment provides a complementary qualitative risk view, enabling more holistic early warning signals for portfolio risk management.

Index Terms— Stock risk management, Time series, Logistic regression, Value at Risk, Sharpe ratio, News sentiment, Gemini API

1. INTRODUCTION

Financial institutions and retail investors as well as the company itself increasingly rely on systematic methods to monitor and manage the risk associated with individual stocks. Large drawdowns, regime shifts are often driven by both price dynamics and news flows. Traditional risk metrics such as volatility or Value at Risk (VaR) capture price-based risk but ignore qualitative information from news, whereas pure sentiment systems lack explicit links to market risk factors. An effective early warning system must combine both perspectives while being robust to the statistical pitfalls of financial time series.

This project focuses on the construction of a stock-level early warning risk management tool that integrates (i) internal market data and (ii) external qualitative information of companies based on news articles. The goal is to classify each trading day into low, medium or high risk and to expose these risk levels through an interactive dashboard that can support practitioners in monitoring exposures.

From a machine learning perspective, financial time series present challenges such as non-stationarity, autocorrelation and the danger of future information leaking into the training process. To address these issues, the system adopts a strictly time-based train-test split and uses expanding-window statistics so that only past information is used to compute features at each point in time. On the NLP side, the Gemini API is prompted to summarize recent news about a company and produce a structured JSON describing sentiment, key risks, and qualitative risk level.

Section 2 formulates the problem and the specific objectives of the internship project. Section 3 describes the methodology, including feature engineering, risk labelling and the integration of Gemini-based sentiment. Section 4 presents the experimental setup and quantitative results obtained on Amazon stock. Section 5 discusses the implications and limitations of the approach. Section 6 concludes with a summary and directions for future work, and Section 7 lists relevant artifacts and demonstrations. Section 8 presents the references used for this project.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1. Problem Statement

The problem addressed in this project is to design and implement an early warning system for stock-level risk that combines market-based quantitative indicators with qualitative news sentiment. Given a historical time series of daily price data for a stock, along with streaming public news about the corresponding company, the task is to estimate on each day a discrete risk level (low, medium, high) and associated risk probabilities that can be used to trigger alerts.

Several constraints shape the formulation:

- **Time-series integrity:** The model must not use any information from the future to compute features or make predictions for a given day, thereby avoiding look-ahead bias.
- **Explainability:** The risk factors should be interpretable to risk managers, motivating the use of volatility, Sharpe ratio and VaR rather than opaque black-box representations.
- **Complementarity of signals:** Both internal market data and external news sentiment should be considered, as idiosyncratic events are often first reflected in news coverage before fully propagating into prices.
- **Practical deployability:** The system should be implementable in a production-friendly stack (Python, scikit-learn, Streamlit) and capable of running on standard hardware.

2.2. Objectives

The specific objectives of the internship project are:

1. To collect and preprocess historical daily stock price data (OHLC) for selected equities (e.g., Amazon) from yfinance.
2. To design time-series-safe risk features, including annualized volatility, 95% Value at Risk, Sharpe ratio, and simple momentum and range indicators, using only past information at each time step.
3. To construct a supervised learning pipeline that maps these features to discrete risk levels via multinomial logistic regression, using a chronological train-test split and strong regularization.
4. To integrate the Gemini Generative Language API for recent news analysis and sentiment extraction, producing a structured qualitative risk summary.
5. To develop an interactive Streamlit dashboard that visualizes risk metrics, model performance and Gemini-based news sentiment in a unified early warning interface.

3. METHODOLOGY / APPROACH

This section describes the overall workflow and the technical details of the proposed system, from data acquisition to model training and dashboard integration.

3.1. Overall Workflow

Figure 1 illustrates the end-to-end workflow. The OHLC data for a target stock are downloaded and cleaned. Time-series-safe risk features are then generated using expanding-window statistics. Based on these features, each day is assigned a risk label (low, medium, high) derived from quantiles and thresholds on volatility, VaR and Sharpe ratio. A multinomial logistic regression model is trained on the earlier portion of the time series and evaluated on a held-out, chronologically later period.

The system also queries the Gemini API with the company name and stock symbol. A carefully engineered prompt instructs Gemini to read recent public news, focus on the latest 7-14 days and respond with a concise JSON object containing the overall sentiment, a short news summary, main risks and positives, and a qualitative risk level. The Streamlit dashboard then presents both the numerical model-based risk classification and the Gemini-based qualitative view to the user.

3.2. Technical Details

3.2.1. Data and Preprocessing

Daily price data (Date, Open, High, Low, Close, Volume) for Amazon (AMZN) from 4 January 2010 to 29 December 2023 were obtained using the Yfinance Python interface and stored as CSV. The Streamlit application loads the CSV, parses the `Date` column as a datetime type, sorts the records chronologically and drops rows with missing values. For Amazon, the cleaned dataset consists of 3522 daily observations and eight columns.

3.2.2. Time-Series-Safe Feature Engineering

A key design choice is to prevent information leakage from the future. To achieve this, all risk features are computed using expanding windows that only aggregate data up to the current day.

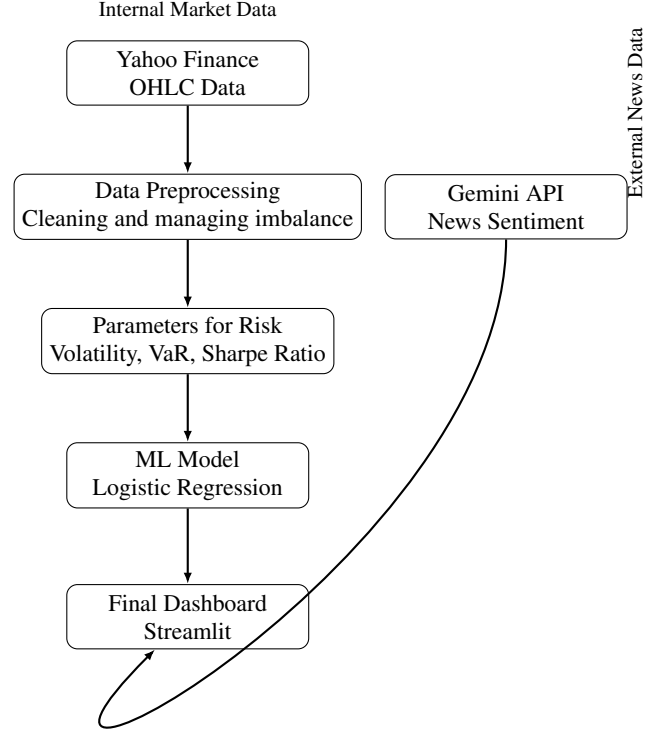


Fig. 1. Overall workflow of the early warning risk management system. Internal market data produce time-series-safe risk features, while external news is analysed via Gemini; both are consolidated in a Streamlit dashboard.

Let P_t denote the closing price at day t . The daily return is defined as

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}. \quad (1)$$

The expanding-window volatility at time t is given by

$$\sigma_t = \sqrt{\frac{1}{t-1} \sum_{i=1}^t (r_i - \bar{r}_t)^2}, \quad (2)$$

where $\bar{r}_t$ is the mean of returns up to t . The annualized volatility assumes 252 trading days:

$$\sigma_t^{\text{ann}} = \sigma_t \sqrt{252}. \quad (3)$$

The 95% Value at Risk (VaR) is approximated non-parametrically using the empirical 5th percentile of the expanding return distribution:

$$\text{VaR}_t^{95\%} = Q_{0.05}(r_1, \dots, r_t), \quad (4)$$

with an annualized version obtained analogously by scaling with $\sqrt{252}$.

The Sharpe ratio at time t uses the average excess return over a risk-free rate r_f (here, a daily risk-free rate of $0.05/252$ is assumed):

$$S_t = \frac{E[r_i - r_f]_{i \leq t}}{\sigma_t}. \quad (5)$$

Additional features include a 5-day momentum

$$\text{Mom}_t^{(5)} = \frac{P_t - P_{t-5}}{P_{t-5}}, \quad (6)$$

and the high–low range normalized by the close,

$$\text{HLRange}_t = \frac{H_t - L_t}{P_t}, \quad (7)$$

where H_t and L_t denote the high and low prices at day t . Rows with insufficient history (e.g., first few days) are dropped after feature computation.

3.2.3. Risk Labelling

To map continuous risk indicators into discrete classes, a rule-based scoring mechanism is employed. For each day, a risk score is computed by aggregating contributions from volatility, VaR, and Sharpe ratio:

- Volatility: if σ_t^{ann} is above the 75th percentile of its historical distribution, add 2 points; if above the 25th percentile, add 1 point.
- VaR: if $\text{VaR}_t^{95\%} < -2\%$ (larger potential loss), add 2 points; if $\text{VaR}_t^{95\%} < -1\%$, add 1 point.
- Sharpe ratio: if $S_t < 0$, add 1 point.

The final risk level is then assigned as:

- Low risk (class 0) if score < 2 ,
- Medium risk (class 1) if $2 \leq \text{score} < 4$,
- High risk (class 2) if score ≥ 4 .

Note: This paper proposes a novel risk scoring mechanism that aggregates distribution-aware volatility quantiles, absolute VaR thresholds, and Sharpe ratio penalties into interpretable daily risk classes for stock prediction.

3.2.4. Model and Training Protocol

A multinomial logistic regression model is used to classify the risk level based on the feature vector

$$\mathbf{x}_t = [\sigma_t^{\text{ann}}, \text{VaR}_t^{95\%}, S_t, \text{Mom}_t^{(5)}, \text{HLRange}_t]. \quad (8)$$

Let $y_t \in \{0, 1, 2\}$ denote the risk label. The model learns class probabilities

$$p(y_t = k | \mathbf{x}_t) = \frac{\exp(\beta_k^\top \tilde{\mathbf{x}}_t)}{\sum_{j=0}^2 \exp(\beta_j^\top \tilde{\mathbf{x}}_t)}, \quad (9)$$

where $\tilde{\mathbf{x}}_t$ is the standardized feature vector and β_k are class-specific parameters. Training minimizes the regularized negative log-likelihood

$$\mathcal{L} = - \sum_t \log p(y_t | \mathbf{x}_t) + \lambda \sum_k \|\beta_k\|_2^2, \quad (10)$$

with λ controlled via the inverse regularization parameter C (here $C = 0.01$ for strong L2 regularization).

Crucially, the train–test split is chronological: 80% of the earliest days are used for training, and the remaining 20% for testing. This mimics real-time deployment, where a model trained on the past is applied to future unseen days.

3.2.5. Gemini API-Based News Sentiment

The external qualitative signal is obtained by calling the Gemini Generative Language API with a structured prompt. The prompt specifies the company name and stock symbol, instructs Gemini to focus on the last 7-14 days of public news, and asks for a JSON-only response with fields:

- `overall_sentiment`: positive, neutral, or negative;
- `news_summary`: 2–3 sentence textual summary;
- `main_risks`: list of short risk bullet points;
- `main_positives`: list of short positive bullet points;
- `qualitative_risk_level`: Low, Medium, or High;
- `time_horizon_comment`: comment on near-term relevance.

4. EXPERIMENTS AND RESULTS

4.1. Experimental Setup

4.1.1. Dataset and Splits

The main experiments were conducted on Amazon (AMZN) daily stock data. After cleaning, the dataset consists of 3522 daily observations spanning from 4 January 2010 to 29 December 2023. The first 80% of the time series (2813 days) form the training set, and the remaining 20% (704 days) constitute the test set. A small buffer of initial days is excluded due to the expanding-window feature computation.

4.1.2. Implementation Details

The system is implemented in Python using:

- `pandas`, `numpy` for data handling and feature engineering,
- `scikit-learn` for scaling and multinomial logistic regression,
- `plotly` and `Streamlit` for interactive visualization and deployment,
- `requests` for calling the Gemini REST API.

All experiments were run on a standard laptop with no GPU requirements, confirming that the pipeline is lightweight.

4.1.3. Evaluation Metrics

Model performance is evaluated on the test set using standard multi-class metrics:

- **Accuracy**: overall fraction of correctly classified days.
- **Precision (weighted)**: average precision across classes, weighted by class support.
- **Recall (weighted)**: average recall across classes, weighted by class support.
- **F1-score (weighted)**: harmonic mean of precision and recall, weighted by support.

4.2. Results

For Amazon, the pipeline produced the following key descriptive statistics on the full period:

- Mean annualized volatility: approximately 32.6%.
- Mean Sharpe ratio: approximately 0.04, indicating modest excess return relative to volatility.

Table 1. Performance of multinomial logistic regression on Amazon test period (704 days).

Metric	Accuracy	Precision	Recall	F1-score
Value	0.808	0.918	0.808	0.835

- Mean annualized 95% VaR: approximately -46.0% , reflecting the tail risk captured by the empirical expanding-window percentile.

Table 1 summarizes the classification performance on the held-out test set.

The confusion matrix shows that most misclassifications occur between adjacent risk levels, which is expected given the continuous nature of the underlying risk factors. High-risk days are less frequent but are generally identified with reasonable precision due to the volatility or VaR thresholding used for labelling.

The dashboard also provides time-series plots of volatility, VaR, and Sharpe ratio, illustrating how these metrics evolve over the 2010-2023 period, as well as a bar chart of the distribution of risk labels (low, medium, high).

The Gemini-based news sentiment module produces, for a given date range, an overall sentiment label (e.g., neutral or negative), short lists of main risks and positives, and a qualitative risk level tag. While this external signal is not yet directly integrated as a model feature, it serves as an important qualitative overlay for interpreting the numeric risk outputs.

5. DISCUSSION

The results indicate that a relatively simple, interpretable model—multinomial logistic regression on carefully engineered time-series-safe features—can achieve strong performance in classifying daily risk levels for a large-cap stock like Amazon. The weighted precision of 0.918 and F1-score of 0.835 suggest that the model is not only accurate overall but also handles class imbalance reasonably well.

One reason for this performance is the use of expanding-window statistics, which impose a realistic information structure: at any time t , only data from days $\leq t$ are used. This avoids look-ahead bias that often inflates performance in financial ML experiments. Additionally, the risk labelling scheme, driven by quantiles of volatility and VaR as well as the sign of the Sharpe ratio, embeds domain knowledge about what constitutes “high” risk, and thus provides a meaningful learning target.

Logistic regression is a linear model; this simplicity enhances explainability and reduces overfitting, particularly with strong L2 regularization. Also, the risk labels themselves depend on hand-crafted thresholds and quantiles; different choices could shift the class boundaries and alter performance metrics. The Gemini news analysis currently provides qualitative insights—showing users the recent news sentiment and key risk factors—but isn’t part of the numerical logistic regression model.

In the future, negative news sentiment or major company-specific risks (like regulatory probes or earnings shocks) could adjust the model’s risk thresholds or be converted into numerical features for better predictions.

6. CONCLUSION AND FUTURE WORK

This internship project developed a practical early warning risk management system that combines internal market data from Yahoo

Finance with external news sentiment from the Gemini API. By using expanding-window volatility, VaR and Sharpe ratio as features and training a multinomial logistic regression with a time-based train–test split, the system achieves robust and interpretable risk classification for Amazon over the period 2010-2023. The Streamlit dashboard provides an integrated view of risk metrics, model performance and qualitative news-based risk assessment.

Key contributions include: (i) a time-series-safe feature engineering and evaluation pipeline that avoids look-ahead bias; (ii) an interpretable mapping from financial risk factors to discrete risk levels; and (iii) a practical integration of LLM-based news sentiment into a risk monitoring interface.

Future work can proceed in several directions. First, extending the system to a cross-sectional universe of stocks would allow portfolio-level early warning and comparative risk analysis. Second, more expressive models (e.g., gradient boosting, recurrent neural networks) could be investigated, while carefully preserving time-series integrity and explainability. Finally, incorporating more advanced risk measures, such as conditional VaR or drawdown-based indicators, and performing backtesting of alert strategies would move the system closer to real-world deployment.

7. ARTIFACTS AND DEMONSTRATIONS

The following artifacts accompany this report and document the implementation:

- **Code Repository:**

<https://github.com/vanshi18s/ML-Based-EWS-for-Risk-Management>

Contains the Streamlit application, feature engineering scripts, and model training code.

- **Interactive Dashboard (Demo):** <https://ml-based-ews-for-risk-management-9jkhilevptlk4yqupstreamlit.app/>

Public demo of the risk management interface, including CSV upload, risk visualizations, and Gemini news sentiment.

- **Video Demonstration:** <https://youtu.be/W4XQPoSZ8-Y?si=59Qd17XhW2OS0--->

Short walkthrough of the system, explaining key components and showcasing Amazon risk results.

- **Internship Certificate:** https://drive.google.com/file/d/1eLMpl7CX-qoGPdQbI9xiv_GMA1Zj8IgG/view?usp=drivesdk

These artifacts are designed to make the project reproducible and to illustrate system behaviour in practice. All external libraries and APIs (including Gemini) are appropriately cited in the accompanying repository documentation.

8. REFERENCES

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